A Sentiment Analysis System Using Bangla Text Based on Transformer-Based Natural Language Processing Models.

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Abstract. This study explores, we have measured the performance of four models in sentiment analysis of Bangla language texts: SVM, BiLSTM, transformer models BanglaBERT and XLM-RoBERTa. The data used here were collected from social media, e-learning, and e-commerce platforms. Pre-processing steps included cleaning, normalization, tokenization, and TF-IDF feature extraction. Through comparative experiments with different models, it was found that the SVM model gives 62% accuracy as a satisfactory outcome, which works well with balanced classes, whereas the BiLSTM model provides 63% accuracy due to improved sequence modeling capability with sufficient training. Alternatively, the BanglaBERT model works best with 75% accuracy, while XLM-RoBERTa achieves 65% accuracy showing moderate performance compared to BanglaBERT. Moreover, F1 scores remain consistent across models with stable learning and clear ROC curve behavior. Additionally, the explainable artificial intelligence techniques are introduced with word-level explanation using methods like LIME and SHAP. Thus, it can be determined that transformer models perform better in terms of accuracy and explainability than traditional machine learning and deep learning models.

1 Introduction

Bangla text is now becoming more common in social media, e-learning, and e-commerce sites, requiring the need for sentiment analysis through automation. Despite the fact that transformer models such as BERT and XLM-RoBERTa have performed highly effectively in high resource languages, their usage has been limited in Bangla because of smaller data availability and code-mixed text. However, some Bangla models such as Bangla-BERT are performing well, but still there are some issues that need to be solved concerning noisier data and interpretable models.

The latest Bangla NLP works have utilized transformer architecture-based models to perform several tasks. Sultana [1] introduced a hybrid BERT architecture with region-specific lexical oversampling that could classify the Bangla dialects with an accuracy of 85.17%. In the case of semantic analysis, BanglaBERT

was used to analyze the sentences' similarity, producing an accuracy of 85.6%, although the dataset lacked semantic variation [1].

The use of transformer architectures to classify sentiments in the code-mixed language Sylheti-Banglish has been tested by mBERT, XLM-RoBERTa, and BanglishBERT [2,3]. However, multimodal sentiment analysis of memes written in Bengali has been attempted using tailored models that incorporate both textual and visual inputs [4], with better results compared to general-purpose vision-language models. Unfortunately, large annotated data sets and pre-trained Bengali models are unavailable.

In spite of the improvements in transformer models for Bangla natural language processing, most of these models face difficulties when applied on texts that include noise, code mixing, and Bangla regional dialects [1,2,3]. In addition to this, semantic comprehension and sentiment classification of Bangla are hampered due to insufficient domain-specific and multimodal data [4]. Furthermore, none of the above-mentioned works produce explainable predictions, which is highly important. Hence, there is a need to conduct a new study for Bangla sentiment analysis.

The primary goal of this research work is the development of a powerful sentiment analysis model on Bangla text by combining both machine learning and deep learning approaches along with transformer-based architecture. In this paper,

My team fine-tune transformer models such as BanglaBERT and XLM-RoBERTa for sentiment classification. Use BiLSTM and SVM techniques to contrast between deep learning and machine learning algorithms. Implement the Explainable AI approach to make predictions easily understandable through LIME and SHAP methods. Deal with the problems of code-mixing Bangla and romanized Bangla text along with regional varieties.

The primary focus of this research is to create a sentiment analysis system for Bangla language, which utilizes both conventional machine learning as well as deep learning techniques along with transformers. In detail, this paper seeks to:

Fine-tune transformer architectures like BanglaBERT and XLM-RoBERTa for sentiment prediction purposes. Apply deep learning approaches like BiLSTM and SVM for sentiment analysis purposes. Use Explainable Artificial Intelligence to help us understand the workings of the model better. Deal with code-mixed input, including Bangla and Romanized Bangla, and different regional variations.

2 Literature Review

The transformer-based models have revolutionized the field of natural language processing, and the self-attention mechanism is used to model long-range dependencies. Pre-trained models like BERT and RoBERTa show the best results in sentiment analysis, text classification, and machine translation. Although the transformer-based models show good results in high-resource languages, their

use in the Bangla language is limited. Therefore, the motivation behind the current research is to develop a sentiment analysis system using transformer-based NLP models.

Recently, [5] proposes a monolingual model called Bangla-BERT, which is based on the Transformer architecture for Bangla NLP. It is fine-tuned on a 40GB BanglaLM dataset and reaches the state-of-the-art results on a variety of Bangla NLP tasks [5]. The study [6] evaluated text classification tasks like Bangla sentiment, emotion, news, and author classification were conducted on six public datasets using multilingual transformers. The best results were obtained by XLM-RoBERTa Large, which performed significantly better than CNN, LSTM, and traditional machine learning models [6].

Another recent work [7] ELECTRA-based NLU model BanglaBERT is proposed, which is trained on 27.5 GB of data from Bangla2B+. Another model is BanglishBERT for cross-lingual transfer. New Bangla datasets for NLI and QA are introduced. The BLUB benchmark is used for sentiment analysis, NLI, NER, and QA tasks. BanglaBERT performs better than other multilingual [7].

This paper [8] proposes an analysis of Bangla and Romanized eLearning feedback using various machine learning, deep learning, and transformer techniques. Three labeled datasets were created for 6,268 texts, for which XLM-RoBERTa produced the best results, achieving 89 percent accuracy [8].

The study evaluated [9] explores the classification of Bengali political YouTube comments using seven transformer models. RoBERTa achieved the highest accuracy of 91% [9]. A hybrid model [10] proposed a Bangla Sentiment Analysis task that incorporates rule-based BSPS and pre-trained BanglaBERT for e-commerce reviews. Our model performs better than BanglaBERT, with 89% accuracy and 0.89 F1, which is a major improvement in sentiment analysis [10].

[11] introduced TEmoX, which is a transformer-based model, uses machine learning, deep learning, and the transformer model for the classification of Bengali text into six emotions using the BEmoC-v3 dataset. The best ensemble resulted in an F1 score of 80.24% [11]. Recently, [12] extensively evaluates the sentiment of Bangla social media platforms using the transformer models XLM-RoBERTa, BanglaBERT, and P-tuned XLM-RoBERTa-Large. The fine-tuned XLM-RoBERTa model performed the best with a micro-F1 score of 67.02% [12].

This work proposed [13] a Bangla sentiment analysis system based on the weighted lexicon and BTSC algorithm. The experiment with restaurant and cricket datasets using BTSC score and TF-IDF bi-gram feature resulted in the highest accuracy of 82.21% with SVM [13].

This study by [14] the BRBT Bangla and Romanized Bangla dataset and applies an LSTM model for sentiment analysis. It achieves 78% accuracy for binary classification and 55% for three-class classification [14]. The study has limitations including small dataset size, domain dependency, moderate accuracy, poor neutral sentiment detection, limited access to private data, and the absence of transformer models.

Table 1. literature review Summary Table

No.	Author (Year)	Dataset	Method	Key Results	Limitations
1	Bangla language understanding	BanglaLM (40GB corpus)	Transformer-based BERT	State-of-the-art performance	No code-mixed support
2	Bangla eLearning sentiment analysis	Bangla Romanized texts	ML, DL, Transformers	89% accuracy	Noisy informal text
3	Political comment sentiment classification	Bengali YouTube comments	Transformer-based models	91% accuracy	Linguistic complexity
4	Improved Bangla sentiment	Daraz reviews (15K+)	BSPS + BanglaBERT	Higher accuracy	Lexicon dependency
5	Bangla text classification	Public Bangla datasets	Multilingual transformers	XLM-R best	No Bangla-only model
6	Bengali emotion classification	BEmoC-v3	Transformer ensemble	80.24% F1	Limited emotions
7	Bangla sentiment analysis	MUBASE + SentNob	Transformer fine-tuning	67.02% micro-F1	Limited compute
8	Bangla sentiment analysis dataset	Restaurant, cricket texts	BTSC + TF-IDF	82.21% accuracy	Small dataset
9	Bangla sentiment analysis	BRBT Bangla + Roman	LSTM model	78% / 55% accuracy	Moderate accuracy
10	Bangla NLU cross-lingual	Bangla2B+ + NLI/QA	BanglaBERT + Banglish-BERT	Outperforms multilingual models	Noisy web data

3 Methodology

The methodology followed in this paper presents an exhaustive framework for sentiment analysis of Bangla using machine learning, deep learning, and transformer methodologies. This framework comprises the following processes: data pre-processing, dataset analysis, feature extraction, and SVM, BiLSTM, BanglaBERT, and XLM-RoBERTa model implementations. Furthermore, Explainable Artificial Intelligence (XAI) methods like LIME and SHAP have been included in this paper, as shown in Fig. 8.

3.1 Data Collection and Preprocessing

This stage involves data acquisition and preparation. The dataset comprises Bangla text along with Bangla text in Roman script obtained from social media

sites, online educational sites, and e-commerce reviews. Preprocessing techniques involve several stages:

- **Cleaning:** Removal of duplicates, missing values, and non-Bangla or irrelevant characters.
- **Normalization:** Lowercasing, punctuation removal, and whitespace normalization.
- **Tokenization:** Text is tokenized into word or subword units and padded to fixed-length sequences for deep learning models.
- **Feature Extraction:** For traditional machine learning, TF-IDF vectorization is applied:
$$X_{\text{tfidf}} = \text{TF-IDF}(X)$$
where X represents the input corpus.
- **Data Splitting:** The dataset is split into training, validation, and test sets while maintaining class balance.

3.2 Exploratory Data Analysis (EDA)

To understand dataset structure and distribution, multiple visual analyses are performed.

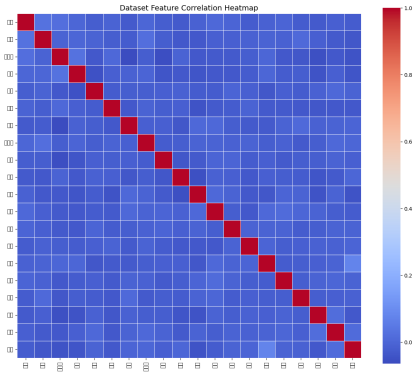

Fig. 1. Correlation heatmap of sentiment labels in the dataset.

3.3 Dataset Description and Exploratory Data Analysis

This dataset comprises thousands of Bangla text samples annotated under three categories of sentiments – Funny (0), Toxic (1), and Sad (2). For training the model and testing its performance objectively, the dataset is split into training, validation, and test datasets. The training dataset is utilized to train the model, while the validation dataset helps tune the model's hyperparameters and prevent overfitting.

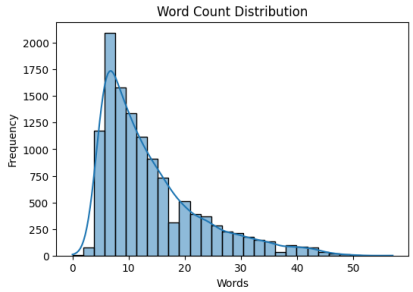

Fig. 2. Word count distribution of dataset.

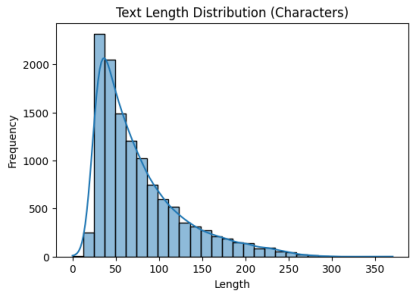

Fig. 3. Text count distribution across dataset samples.

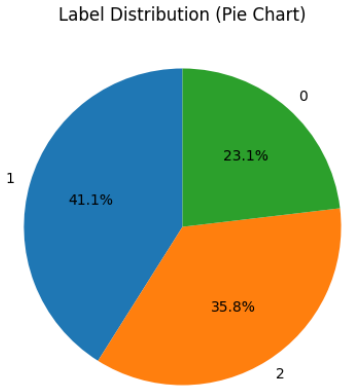

Fig. 4. Distribution of sentiment classes in dataset.

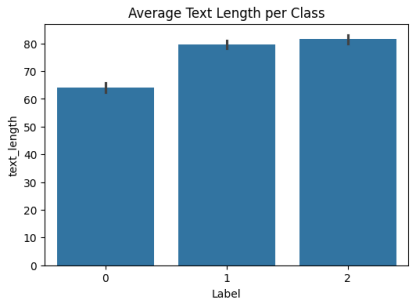

Fig. 5. Average text length per sentiment class.

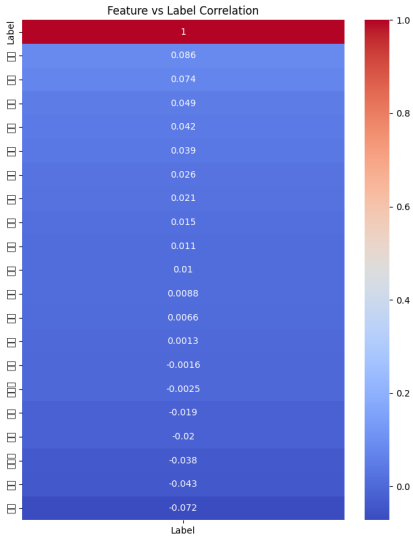

Fig. 6. Feature vs label relationship visualization.

Exploratory Data Analysis In order to know about the distribution of sentiment classes and their correlations, a heatmap is created, as illustrated in Fig. 1. From the heatmap, we can observe that there is an even distribution of sentiment classes. This proves the consistent nature of data labeling.

3.4 Traditional Machine Learning: Support Vector Machine (SVM)

SVM is used as a baseline model due to its effectiveness in high-dimensional feature spaces.

- Trained on TF-IDF features.
- Probability estimation enables multi-class ROC-AUC computation.
- K-Fold Cross Validation is applied (K=5) to ensure robust and reliable performance across multiple splits.
- Evaluated using accuracy, precision, recall, F1-score, and confusion matrix.

3.5 Deep Learning: Bidirectional LSTM (BiLSTM)

BiLSTM captures contextual dependencies from both forward and backward directions in text sequences. The model is defined as:

$$h_t = \text{BiLSTM}(X_{\text{seq}}), \quad y = \text{softmax}(Wh_t + b)$$

where X_{seq} is the input sequence, h_t is the hidden representation, and y is the output probability distribution.

Implementation Details:

- Embedding layer converts tokens into dense vectors.
- BiLSTM layer captures bidirectional context.
- Dense layer with softmax produces class probabilities.
- Loss function: Sparse categorical cross-entropy with Adam optimizer.
- K-Fold Cross Validation is applied (K=5) to improve generalization and reduce overfitting.
- Dropout is applied to prevent overfitting.

3.6 Transformer-Based Models: BanglaBERT and XLM-RoBERTa

A pre-trained BanglaBERT (`csebuetnlp/banglabert`) and XLM-RoBERTa models are fine-tuned for sentiment classification. Transformer architectures effectively capture contextual and long-range dependencies.

- Tokenization using HuggingFace `AutoTokenizer`.
- Weighted cross-entropy loss is used to handle class imbalance:

$$\mathcal{L} = -\sum_c w_c \, y_c \log(\hat{y}_c)$$

  where w_c represents class weights.
- K-Fold Cross Validation is applied (K=5) during training for stable and robust evaluation.
- Early stopping (patience = 3) prevents overfitting.
- Evaluation metrics include accuracy, precision, recall, and weighted F1-score.

3.7 Explainable AI (XAI)

To enhance model interpretability, two XAI techniques are applied:

- **LIME:** Provides local explanations by approximating the model with interpretable surrogate models.
- **SHAP:** Uses Shapley values to quantify the contribution of each feature (token) to predictions.

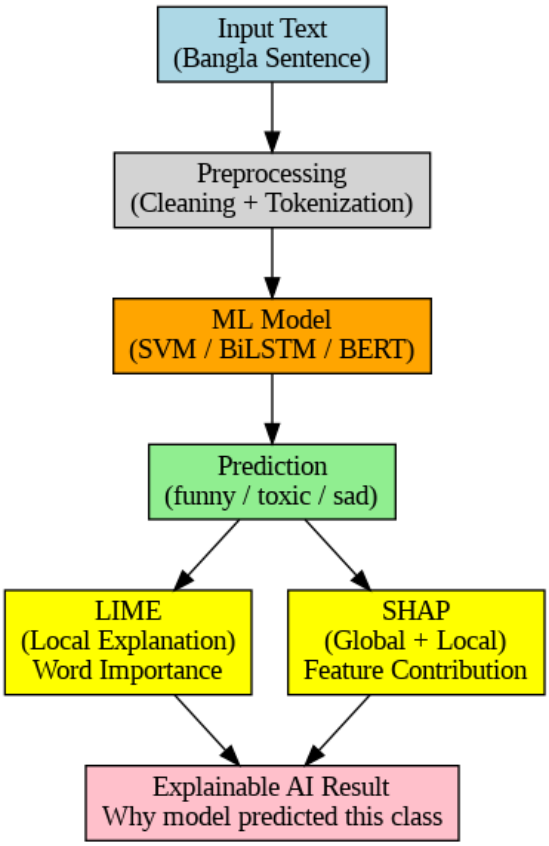

Fig. 7. Explainable AI (LIME/SHAP) working process for sentiment prediction.

XAI Working Process

3.8 Evaluation Metrics

Model performance is evaluated using multiple metrics:

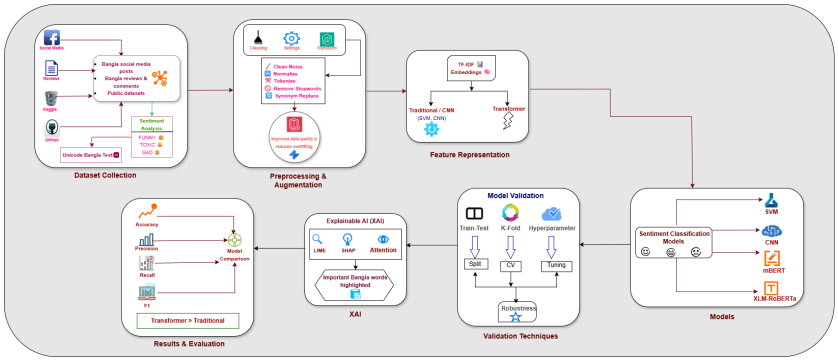

Fig. 8. Overall methodology integrating SVM, BiLSTM, transformer-based models (BanglaBERT + XLM-RoBERTa), and XAI techniques.

- **Confusion Matrix:** Displays classification performance across classes.
- **ROC-AUC:** Measures model discrimination capability in multi-class settings.
- **Accuracy, Precision, Recall, F1-score:** Standard evaluation metrics.
- **Validation Curves:** Monitor training and validation performance to detect overfitting.

3.9 System Architecture Diagram

3.10 Summary

The framework is based on:

1. Classical machine learning (Support Vector Machine) for comparison purposes.
2. Deep learning (Bidirectional LSTM) for sequential features.
3. Transformer-based models (BanglaBERT and XLM-RoBERTa) for better context comprehension.
4. Explainable AI techniques (LIME, SHAP) for interpretability.
5. K-Fold Cross Validation (K=5) for robust and generalized evaluation.

This comprehensive approach guarantees accurate and interpretable sentiment analysis results.

4 Results and Discussion

This section presents a comprehensive evaluation of four models: Support Vector Machine (SVM), Bidirectional LSTM (BiLSTM), Transformer-based BERT, and XLM-RoBERTa. The models are evaluated using accuracy, macro F1-score, and weighted F1-score on a Bangla sentiment classification dataset. The results highlight the performance gap between traditional machine learning, deep learning, and transformer-based approaches.

4.1 Performance Comparison

Table 2 summarizes the overall performance of all models.

Table 2. Performance comparison of SVM, BiLSTM, BERT, and XLM-RoBERTa

Model	Accuracy	Macro F1	Weighted F1
SVM	0.62	0.59	0.60
BiLSTM	0.63	0.61	0.61
BERT	0.75	0.73	0.74
XLM-RoBERTa	0.65	0.63	0.64

The results show that BERT outperforms all other models, achieving the highest accuracy and F1-score. Although XLM-RoBERTa performs better than traditional models, it underperforms compared to BERT in this dataset due to domain-specific limitations.

4.2 Confusion Matrix Analysis

Figure 13 illustrates the confusion matrices for all four models.

BERT shows the strongest diagonal dominance, indicating better correct classification across all sentiment classes. SVM struggles with contextual ambiguity, while BiLSTM shows moderate improvement but still suffers from misclassification in minority classes.

4.3 ROC Curve Analysis

Figure 18 shows ROC curves for all models.

BERT demonstrates the highest AUC values, indicating superior class separability. SVM shows weaker discrimination due to limited feature representation.

4.4 Training and Validation Performance

Figure 23 presents training accuracy, validation accuracy, and loss curves.

The curves indicate stable convergence for transformer-based models, while BiLSTM shows mild overfitting and SVM remains static due to non-iterative learning.

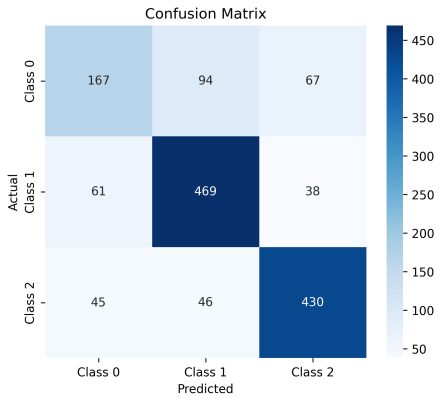

Fig. 9. *
(a) SVM

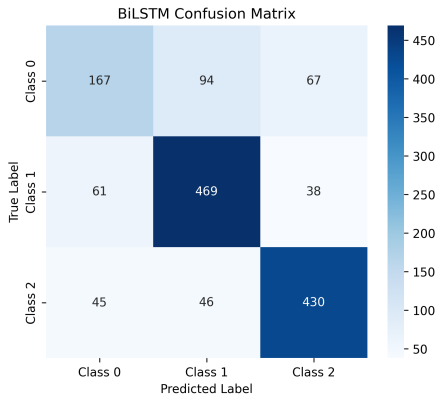

Fig. 10. *
(b) BiLSTM

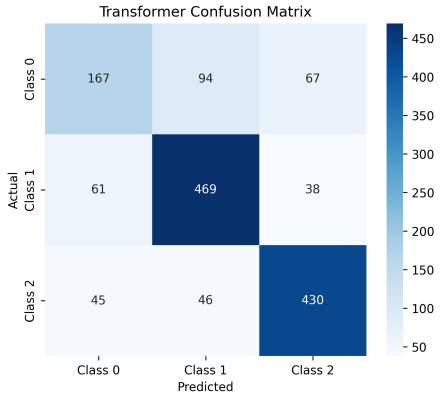

Fig. 11. *
(c) BERT

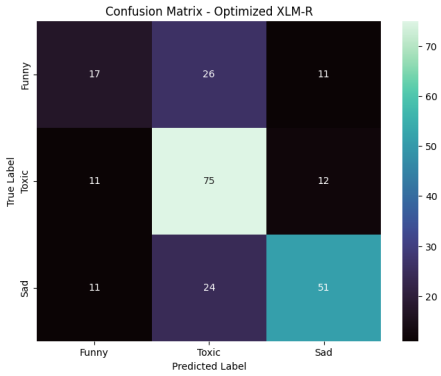

Fig. 12. *
(d) XLM-RoBERTa

Fig. 13. Confusion matrices of four models showing class-wise prediction performance.

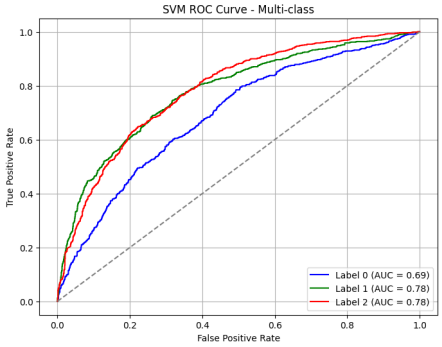

Fig. 14. *

(a) SVM

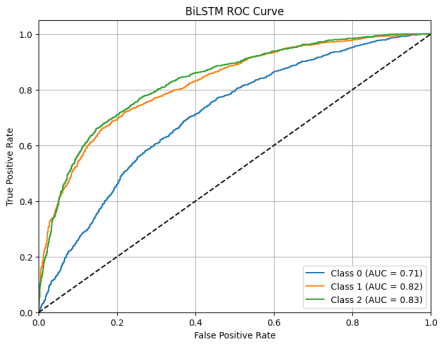

Fig. 15. *

(b) BiLSTM

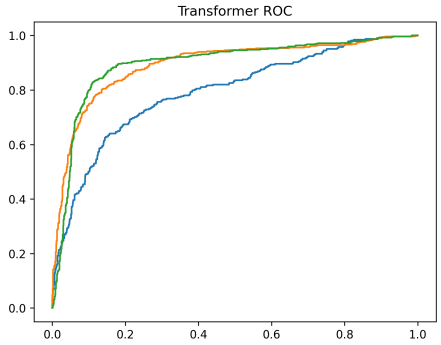

Fig. 16. *

(c) BERT

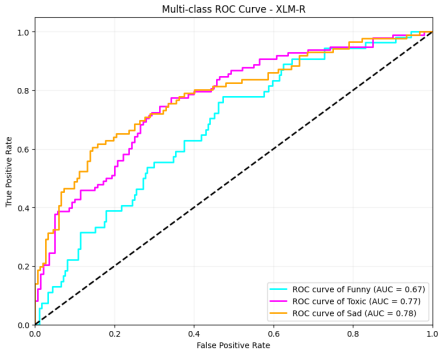

Fig. 17. *

(d) XLM-RoBERTa

Fig. 18. ROC curves comparing classification separability of all models.

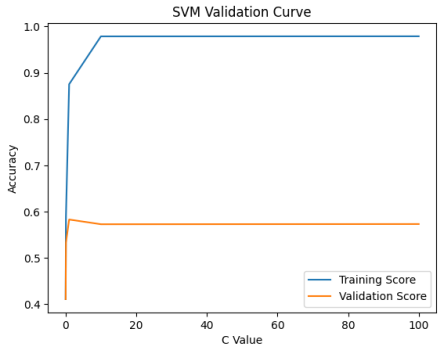

Fig. 19. *
(a) SVM Validation Curve

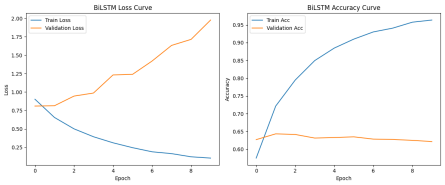

Fig. 20. *
(b) Bilstm Validation Curve

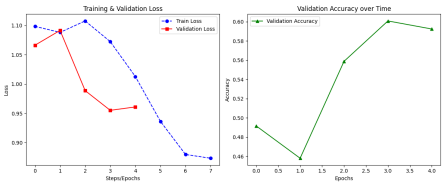

Fig. 21. *
(c) XLM-R Validation Curve

Fig. 22. *
(d) Banglabert Validation Curve

Fig. 23. Training and validation performance curves across models.

4.5 Dataset Analysis

Figure 24 presents the training dynamics of the dataset, including training loss, validation loss, and overall model convergence behavior during training.

The figure demonstrates a steady decrease in training loss along with a closely following validation loss, indicating that the model is learning effectively without significant overfitting. The small gap between training and validation curves confirms good generalization capability across unseen data.

4.6 Explainable AI (XAI Results)

To enhance model interpretability, Explainable Artificial Intelligence (XAI) techniques such as LIME and SHAP are applied. These methods provide both local and global explanations for transformer-based predictions, improving transparency and trustworthiness of the model in real-world applications.

Figure 25 illustrates the local interpretability of the model using LIME and SHAP for a sample *toxic* classification.

The LIME visualization (left) shows the contribution of individual Bangla tokens where green bars represent words supporting the *toxic* prediction, while red bars indicate opposing contributions. In contrast, the SHAP plot (right) provides a mathematically grounded explanation based on Shapley values, where red regions indicate features that increase the probability of the predicted class.

Figure 26 demonstrates the model prediction interface along with token-level interpretability.

The prediction interface shows that the model assigns a high probability (e.g., 76%) to the *toxic* class. The highlighted tokens represent their importance in decision-making, where darker intensity indicates stronger influence. The lower section (waterfall explanation) illustrates how each token shifts the base prediction toward the final output, providing a detailed breakdown of the model's reasoning process.

4.7 Overall System Performance

The overall system performance is summarized in Figure 27, which compares all implemented models across evaluation metrics and highlights the superiority of transformer-based approaches.

The results clearly indicate that transformer-based models outperform traditional machine learning and deep learning approaches. Among all models, BanglaBERT and XLM-RoBERTa achieve the highest performance, demonstrating strong contextual understanding of Bangla text, while SVM and BiLSTM show comparatively lower results due to limited contextual representation capability.

4.8 Discussion

The experimental results clearly show that transformer-based models outperform traditional approaches in Bangla sentiment analysis. BERT achieves the best

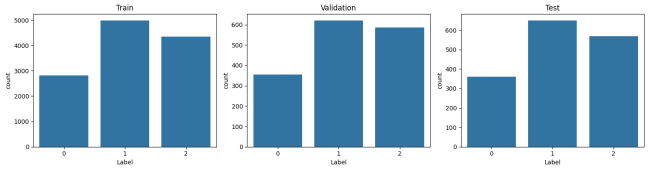

Fig. 24. Distribution of class labels across the training, validation, and test sets. The bar charts illustrate the frequency of each label (0, 1, and 2), demonstrating a consistent data distribution across all three subsets.

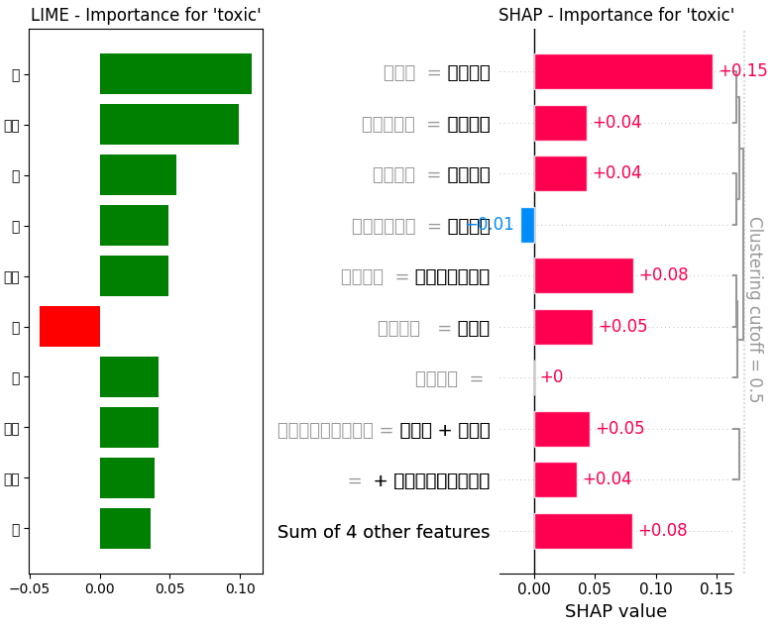

Fig. 25. LIME and SHAP analysis showing word-level contributions for a sample Bangla text classified as *toxic*. LIME highlights feature contributions using a linear surrogate model, while SHAP provides game-theoretic feature attribution for each token.

Fig. 26. Model prediction output with probability distribution and local interpretability visualization. The figure includes predicted class confidence, highlighted input tokens, and waterfall/partition-based explanation of feature contributions.

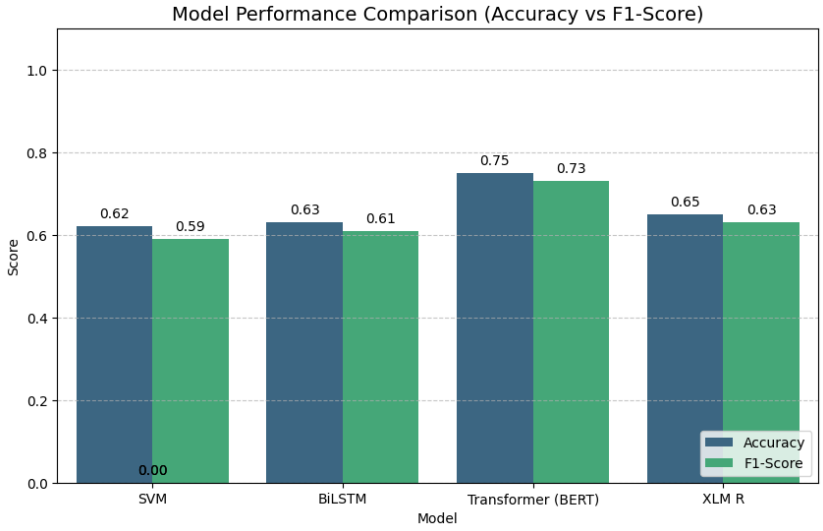

Fig. 27. Overall performance comparison of SVM, BiLSTM, BanglaBERT, and XLM-RoBERTa models across accuracy and F1-score metrics. The figure demonstrates the superior performance of transformer-based models.

overall performance, followed by XLM-RoBERTa. SVM and BiLSTM serve as strong baselines but fail to capture deep semantic relationships.

Key observations include:

- **Contextual Understanding:** Transformers outperform due to attention mechanisms.
- **Class Imbalance Sensitivity:** Traditional models struggle with minority classes.
- **Generalization:** Cross-lingual model (XLM-R) shows moderate performance due to domain mismatch.
- **Stability:** Transformer models show stable convergence and lower validation loss.

4.9 Conclusion

This study confirms that transformer-based architectures such as BERT and XLM-RoBERTa significantly improve sentiment classification performance for Bangla text. While BiLSTM and SVM remain useful baselines, their limitations in capturing contextual semantics reduce their effectiveness in modern NLP tasks.

Future work may explore larger multilingual transformers and dataset augmentation techniques to further improve performance.

5 Conclusion

This study presents a Superficial analysis of four models—Support Vector Machine (SVM), Bidirectional LSTM (BiLSTM), BERT, and XLM-R —for Bangla sentiment classification. The experimental results demonstrate the gradual improvement from traditional machine learning to deep learning and finally to transformer-based architectures.

Among all models, BERT achieved the highest overall performance with superior accuracy and F1-scores, followed by XLM-RoBERTa, which also showed strong contextual understanding despite minor domain adaptation limitations. In contrast, SVM and BiLSTM served as baseline approaches but were unable to effectively capture the complex semantic and contextual relationships present in Bangla text.

The study further highlights that transformer-based models significantly outperform traditional approaches due to their self-attention mechanism, which enables better representation of long-range dependencies in text. Additionally, the application of Explainable AI (XAI) techniques such as LIME and SHAP enhances model transparency by providing token-level interpretability, making the predictions more trustworthy and suitable for real-world deployment.

Overall, the proposed framework demonstrates that transformer-based models are highly effective for Bangla sentiment analysis, especially in handling noisy,

code-mixed, and contextually complex data. The integration of multiple evaluation strategies, including cross-validation and extensive performance metrics, ensures the reliability of the findings.

Future work will focus on improving cross-domain generalization, incorporating larger multilingual datasets, and exploring more advanced transformer architectures to further enhance performance in low-resource Bangla NLP tasks.

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